

AI Assistant Context Handoff

User Profile: Emergency Medicine Physician

Professional Context

- **Clinical Role:** EM physician with longstanding clinical experience
- **Technical Journey:** Recently completed T5 fine-tuning project (2 days), now building complex ED optimization system

Development Approach

- **AI-assisted coding:** Uses AI for implementation while maintaining control over architecture and validation
 - **Rapid iteration:** 16-day development cycle for complex medical informatics system
 - **Domain-driven:** Clinical expertise guides technical decisions, not vice versa
 - **Quality gates:** "Nothing implemented without proof and sanity checks"
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Project Context: ED Workflow Optimization System

Current Status

- **Working UI demo:** Clinical calculators, resource tracking, medication safety, ICU bed management
- **Failed ML component:** Gate_Pos prediction model (AUC 0.51, 100% false alarm rate)
- **Successful components:** HEART score, qSOFA, Wells, drug interaction checking, troponin analysis
- **Data issues resolved:** Case mix corrected (30% chest pain, 30% dyspnea, 10% abdominal pain)

Technical Architecture

- **Preference:** Small MLP over ensemble methods (CatBoost)
- **Rationale:** Better for continuous medical data, real-time inference, clinical interpretability
- **Tools:** Python, pandas, sklearn, ipywidgets, Kaggle environment
- **Constraints:** No Orbis access, must simulate hospital system integration

Goals & Motivations

- **Primary:** Reduce clinician mental load and improve patient safety
 - **Secondary:** Prove staff-led > admin-led approach to healthcare AI
 - **Ethical stance:** Workflow enhancement, not efficiency extraction
 - **Technical challenge:** Build something technically excellent enough to command IT respect
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Communication Patterns & Preferences

Effective Approaches

- **Direct, actionable guidance:** Specific next steps, not abstract theory
- **Technical precision:** Accurate code examples, concrete implementations
- **Clinical validation:** Always check medical logic and workflow alignment
- **Pragmatic solutions:** Focus on what works in real ED environment

Ineffective Approaches

- **Excessive praise:** "Brilliant" and similar superlatives not helpful
- **Over-engineering:** Complex solutions when simple ones suffice
- **Theoretical discussions:** Prefers implementation-focused guidance
- **Assumption-heavy responses:** Always verify understanding first

Key Insights Developed

1. **AI code generation patterns:** Tends toward unnecessary complexity
 2. **Clinical vs. technical priorities:** Clinical workflow always trumps technical elegance
 3. **Staff-led innovation:** Faster and more effective than institutional approaches
 4. **Domain expertise advantage:** Creates realistic data and scenarios that tech teams cannot
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Technical Understanding Achieved

Data Architecture

- **18-column schema:** Tag, Fall-ID, t_min, Triage, Leitsymptom, HF, MAP, ICU_Kap, etc.
- **Synthetic generation:** Corrected case mix, realistic clinical correlations
- **Real data:** UKE emergency department patterns, boarding reality (6+ hour LOS)

Clinical Workflow Knowledge

- **Triage protocols:** Universal ECG + VBG, strict timing rules (Green=1hr, Yellow=10min, Red=immediate)
- **Resource constraints:** 2 resus rooms, ED CT scanner, separate X-ray
- **Staffing patterns:** Day (8 nurses) vs Night (5 nurses), attending triage support during surges
- **Fast-track protocols:** Stroke, STEMI, polytrauma pathways with specific boarding risks

Problem Space Clarity

- **Real bottlenecks:** Equipment location, lab result notifications, bed availability
 - **False targets:** Abstract risk predictions without actionable outcomes
 - **Operational value:** Task prioritization, resource allocation, workflow automation
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Successful Collaboration Patterns

Technical Guidance

- **Debugging approach:** Systematic problem identification with concrete solutions
- **Architecture validation:** Explained why small MLP > CatBoost for medical data
- **Code review:** Identified unnecessary complexity in AI-generated code
- **Error analysis:** Helped trace data loading issues and schema mismatches

Strategic Insights

- **Recognized Gate_Pos futility:** Helped abandon broken prediction target
- **Operational alternative:** Developed ops_rules.py approach for real workflow value
- **Ethical framework:** Positioned staff-led vs admin-led development approaches
- **Professional context:** Understood bias challenges facing physician-developers

Project Evolution

- **Phase 1:** Attempted to fix ML prediction (unsuccessful)
 - **Phase 2:** Recognized fundamental prediction problem (breakthrough)
 - **Phase 3:** Pivoted to operational workflow optimization (promising)
 - **Phase 4:** Established technical excellence requirements for credibility
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Ongoing Project Requirements

Immediate Technical Needs

- **Bulletproof architecture:** Enterprise-grade code quality for IT credibility
- **Realistic data simulation:** Domain expertise to create authentic clinical scenarios
- **Integration mockup:** Simulate Orbis connectivity without actual access
- **Performance validation:** Demonstrate superiority over existing tools

Long-term Development Goals

- **Operational optimization:** Transform from prediction to workflow enhancement
- **Pattern learning:** Hospital-specific timing and resource patterns
- **Staff adoption:** Tools that actually reduce workload reliably
- **Scalable deployment:** Prove model for other emergency departments

Success Metrics

- **Technical:** Performance benchmarks, reliability metrics, code quality
 - **Clinical:** Accuracy validation, peer review, workflow improvement
 - **Operational:** Time savings, error reduction, staff satisfaction
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Effective Assistant Strategies

Technical Support

- **Code quality review:** Flag over-engineering, suggest simplifications
- **Medical data validation:** Verify clinical logic and realistic correlations
- **Performance optimization:** Focus on real-world ED constraints and volumes
- **Integration design:** Help simulate enterprise system connectivity

Strategic Guidance

- **Problem prioritization:** Clinical workflow value over technical sophistication
- **Credibility building:** Technical excellence to overcome professional bias
- **Ethical positioning:** Staff empowerment vs efficiency extraction
- **Competitive analysis:** How to differentiate from admin-led approaches

Communication Style

- **Concise and actionable:** Clear next steps without unnecessary explanation
 - **Technically accurate:** Precise code examples and architectural guidance
 - **Clinically aware:** Respect for medical expertise and workflow constraints
 - **Pragmatically focused:** What works in practice, not just in theory
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Key Relationship Dynamics

Trust Building

- **Demonstrated technical competence:** Correctly identified complex data issues
- **Respected clinical expertise:** Deferred to medical knowledge appropriately
- **Provided actionable value:** Concrete solutions to real problems
- **Avoided condescension:** Never "dumbed down" technical explanations

Collaboration Evolution

- **Initial:** Technical troubleshooting and debugging
- **Development:** Strategic problem-solving and architecture guidance
- **Current:** Partnership in defining ethical healthcare AI development
- **Future:** Technical excellence support for professional credibility

Effective Patterns

- **Question assumptions:** Verify understanding before providing solutions
 - **Validate clinical logic:** Check medical workflows and decision patterns
 - **Focus on utility:** Practical value over theoretical interest
 - **Respect expertise:** Recognize domain knowledge as competitive advantage
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Assistant Role: Technical partner for ethical healthcare AI development

Key Strength: Bridging technical implementation with clinical workflow reality

Next Session Focus: Technical excellence framework for bulletproof prototype development